

Intro to Data-Driven Dynamics

Aim: Use data to make qualitative and/or quantitative predictions about dynamical systems - systems evolving in time.

Recap

A linear dynamical system has form

$$\dot{x} = Ax, \quad x(0) = x_0.$$

The solution is the matrix exponential

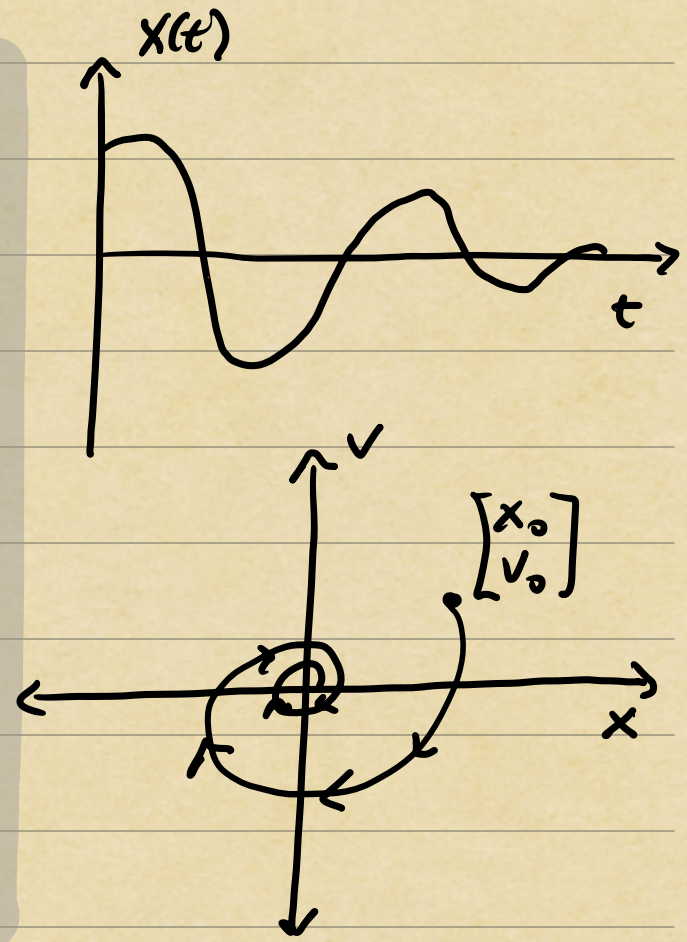
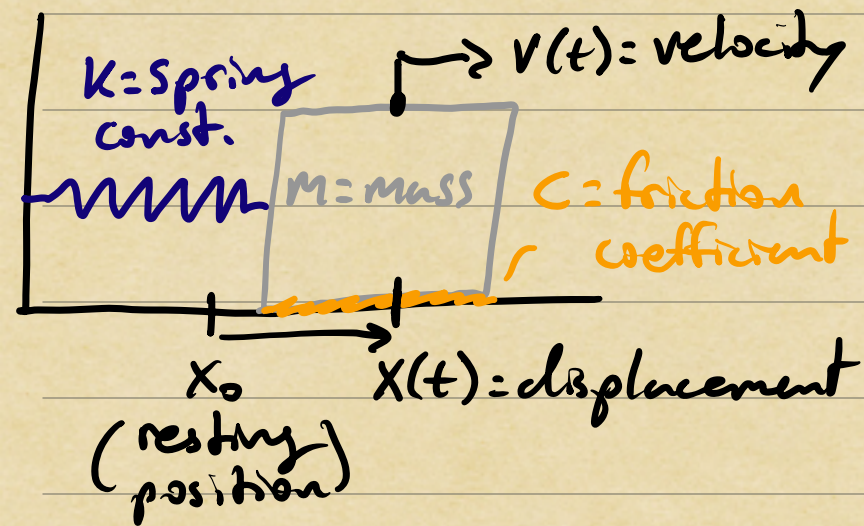
$$x(t) = \exp(At)x_0, \quad t \geq 0.$$

If $Av_j = \lambda_j v_j$, and $\{v_1, \dots, v_d\}$ are a basis,

$$\exp(At) = \underbrace{\begin{bmatrix} | & & | \\ v_1 & \dots & v_d \\ | & & | \end{bmatrix}}_V \underbrace{\begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_d t} \end{bmatrix}}_{e^{-\Lambda t}} \underbrace{\begin{bmatrix} | & & | \\ v_1 & \dots & v_d \\ | & & | \end{bmatrix}^{-1}}_{V^{-1}}.$$

Example. Simple Oscillator

$$\frac{d}{dt} \begin{bmatrix} x \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$



Phase Plane / State Space

The state-space trajectories - which describe the physical behavior of the oscillator - can be understood by looking at the eigenvalues (eigenvectors) of the matrix:

$$\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix} = \lambda \begin{bmatrix} x \\ v \end{bmatrix}$$

Letting $\omega_0 = \sqrt{\frac{k}{m}}$ and $\gamma = \frac{c}{2m}$, then

$$\lambda_{\pm} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}.$$

If $u_+ = \begin{bmatrix} x_+ \\ v_+ \end{bmatrix}$ and $u_- = \begin{bmatrix} x_- \\ v_- \end{bmatrix}$ are the associated eigenvectors, then

$$x(t) = c_+ x_+ e^{\lambda_+ t} + c_- x_- e^{\lambda_- t}$$

$$v(t) = c_+ v_+ e^{\lambda_+ t} + c_- v_- e^{\lambda_- t}$$

where $\begin{bmatrix} c_+ \\ c_- \end{bmatrix} = \begin{bmatrix} x_+ & x_- \\ v_+ & v_- \end{bmatrix}^{-1} \begin{bmatrix} x_0 \\ v_0 \end{bmatrix}.$

The dynamics of the oscillator can be neatly classified into three distinct regimes, depending on the sign of $\gamma^2 - \omega_0^2$.

Case 1: "underdamped" $\gamma < \omega_0$

When $\gamma < \omega_0$, $\text{Im}(\lambda_{\pm}) \neq 0$ and the solutions are oscillatory. Writing

$$\lambda_{\pm} = -\gamma \pm i\omega, \quad \omega = \sqrt{\omega_0^2 - \gamma^2},$$

$$x(t) = e^{-\gamma t} (c_+ x_+ e^{i\omega t} + c_- x_- e^{-i\omega t})$$

$$= A e^{-\gamma t} \cos(\omega t + \varphi)$$

where A (amplitude) and φ (phase) depend on the initial conditions: eq. (1).

Note that the **real part** of the eigenvalues controls the **decay rate** of the amplitude, while the **imaginary part** determines the **frequency of oscillation** around equilibrium.

Case 2: "Critically damped" $\gamma = \omega_0$

When $\gamma = \omega_0$, there is one real eigenvalue

$$\lambda_+ = \lambda_- = -\gamma,$$

with a single lin. indep. eigenvector:

$$x(t) = c_0 e^{-\gamma t} + c_1 t e^{-\gamma t}$$

Note that the solution doesn't oscillate.

Case 3: "Overdamped" $\gamma > \omega_0$

When $\gamma > \omega_0$, there are two distinct real eigs,

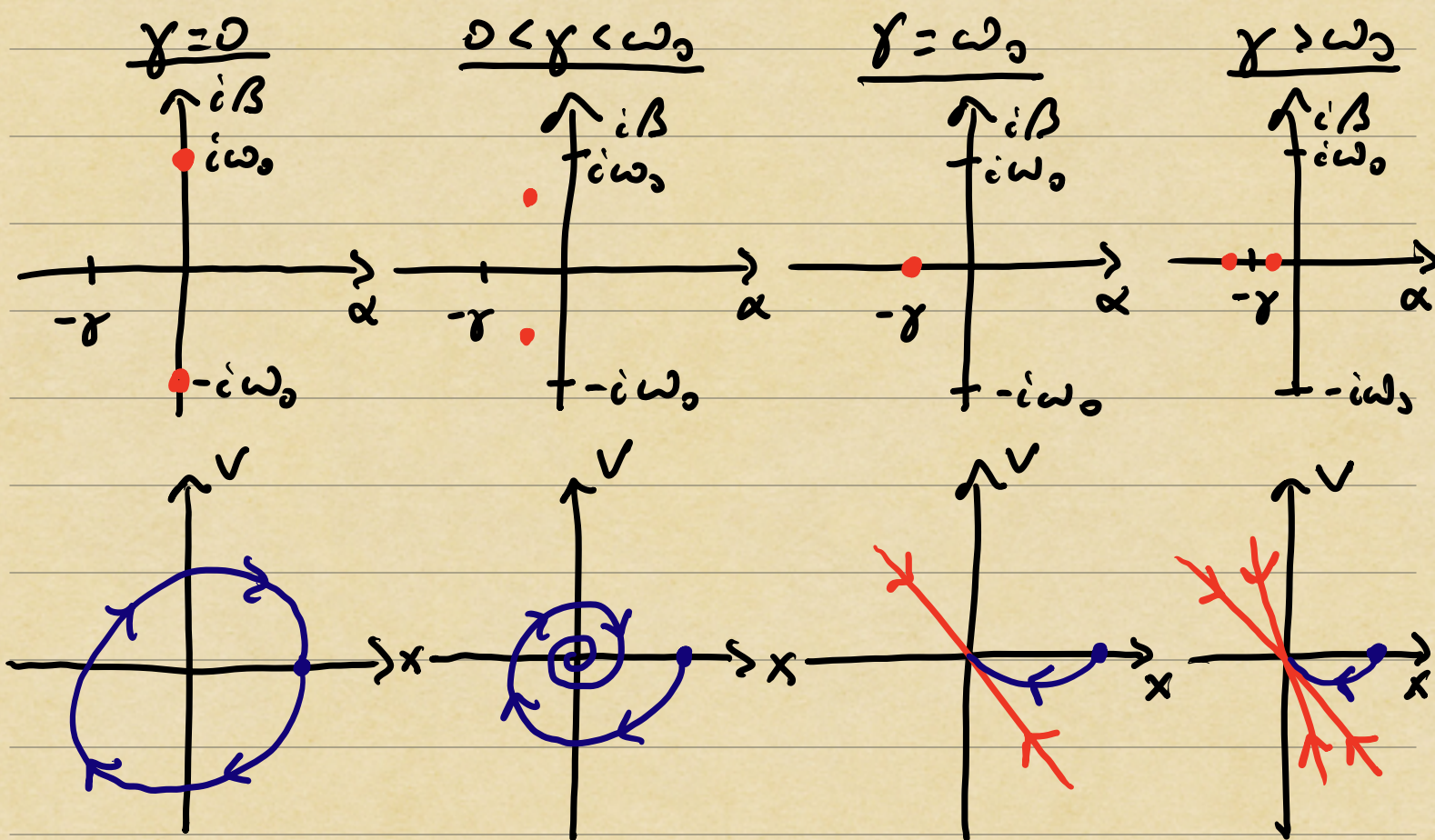
$$\lambda_{\pm} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2},$$

with two linearly indep. eigenvectors:

$$x(t) = C_+ x_+ e^{-\lambda_+ t} + C_- x_- e^{-\lambda_- t}$$

The solutions decay rapidly to the "resting" position of the oscillator.

If we imagine slowly turning the damping coefficient on from $\gamma=0 \rightarrow 2\omega_0$, we can watch the eigenvalues move through the complex plane and observe the transitions in state-space trajectories.



The red lines for $\gamma \geq \omega_0$ represent the real eigenvector(s), along which state-space trajectories approach the origin as $t \rightarrow \infty$, since (e.g., for $\gamma > \omega_0$)

$$\begin{bmatrix} x(t) \\ v(t) \end{bmatrix} = c_+ \begin{bmatrix} x_+ \\ v_+ \end{bmatrix} e^{\lambda_+ t}$$

Q: Can we learn the oscillator's dynamics from trajectory/snapshot data?

Dynamic Mode Decomposition

Suppose that we have snapshot data from a linear dynamical system:

$$X = \begin{bmatrix} | & & | \\ x_1 & \dots & x_m \\ | & & | \end{bmatrix}, \quad Y = \begin{bmatrix} | & & | \\ y_1 & \dots & y_m \\ | & & | \end{bmatrix}$$

where $y_j = \underbrace{F^{\Delta t}}_{\text{"snapshots"}}(x_j), \quad j=1, \dots, m.$

Recall that the map $F^{\Delta t}: \mathbb{R}^d \rightarrow \mathbb{R}^d$ has the matrix exponential form

$$F^{\Delta t}(x_j) = \exp(\Delta t A) x_j$$

for a continuous time system $\dot{x} = Ax$.

Since $F^{\Delta t}: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is linear, one simple/natural idea is to perform a linear fit to the data.

We want to find $\hat{F}^{At} \in \mathbb{R}^{d \times d}$ s.t.

$$\|Y - F^{At}X\|$$

is as small as possible. If $m > d$, then this is an overdetermined least-squares problem:

$$\hat{F}^{At} = \underset{G \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \|X^T G - Y^T\|$$
$${}^m \begin{bmatrix} \\ \\ \\ \end{bmatrix}_d \begin{bmatrix} \\ \\ \\ \end{bmatrix}_m - \begin{bmatrix} \\ \\ \\ \end{bmatrix}_m$$